

BUFFALO 13 ESE, SD, USA

WMO: 726640

Lat: 45.5160N

Lon: 103.3017W

Elev: 879

StdP: 91.20

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94081

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.2	-22.7	-29.5	0.3	-26.5	-25.9	0.4	-22.8	9.9	-1.6	8.7	-3.8	1.4	N/A	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	35.0	18.6	32.9	18.2	30.9	17.8	21.4	29.3	20.4	28.4	19.4	27.4	3.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	15.3	25.6	17.7	14.1	24.2	16.7	13.2	23.0	66.5	29.3	62.6	28.5	59.1	27.3	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.7	6.7	DB	-33.0	38.5	4.0	2.9	-35.9	40.6	-38.2	42.3	-40.5	44.0	-43.4	46.1	(4)
(5)			WB	-33.4	23.1	4.7	1.3	-36.8	24.0	-39.5	24.7	-42.1	25.5	-45.5	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-7.0	-7.3	0.0	5.5	11.3	17.4	21.9	20.1	14.9	6.7	-0.1	-6.7	(6)	
	DBStd	11.83	7.45	8.10	7.29	5.58	4.90	4.06	3.66	3.77	4.83	5.35	6.24	7.18	(7)	
	HDD10.0	2488	527	484	314	153	43	2	0	0	11	129	306	519	(8)	
	HDD18.3	4594	786	717	568	384	224	63	9	26	123	362	554	777	(9)	
	CDD10.0	1197	0	0	4	18	82	223	370	312	159	26	2	0	(10)	
	CDD18.3	263	0	0	0	0	4	36	121	80	21	1	0	0	(11)	
	CDH23.3	4120	0	0	4	21	103	539	1697	1218	495	43	0	0	(12)	
	CDH26.7	1801	0	0	0	4	22	185	832	553	193	12	0	0	(13)	
(14) Wind	WSAvg	2.8	2.8	2.7	3.1	3.4	3.2	2.8	2.5	2.4	2.5	2.7	2.5	2.6	(14)	
(15) Precipitation	PrecAvg	382	11	11	21	36	59	77	55	32	27	32	12	9	(15)	
	PrecMax	526	44	28	58	70	161	133	162	73	100	131	59	25	(16)	
	PrecMin	202	2	1	9	7	18	41	8	5	3	5	0	1	(17)	
	PrecStd	82	9	7	11	20	40	28	33	19	24	32	14	7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	15.8	23.5	25.8	29.6	34.5	39.5	36.4	34.0	28.0	20.8	13.5	(19)	
		MCWB	4.9	7.2	10.7	12.4	16.8	18.9	17.7	18.1	17.6	14.5	9.7	6.1	(20)	
	2%	DB	7.7	10.2	18.9	22.1	26.1	31.1	35.9	34.3	31.3	23.4	16.0	8.4	(21)	
		MCWB	3.1	4.5	8.2	11.5	15.6	18.3	19.2	17.7	16.3	12.4	7.5	3.1	(22)	
	5%	DB	4.9	6.3	15.1	18.8	23.8	28.6	33.6	32.2	29.1	19.8	12.9	5.6	(23)	
		MCWB	1.3	2.3	6.8	9.6	14.0	17.7	18.8	18.1	15.6	10.4	6.1	1.5	(24)	
	10%	DB	3.1	3.5	11.3	16.0	21.2	26.4	31.3	30.0	26.0	16.7	9.8	3.1	(25)	
		MCWB	0.5	0.8	5.3	8.2	13.3	16.9	18.5	17.7	14.8	9.2	4.4	0.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	7.9	11.2	13.5	18.6	22.2	23.4	21.5	20.1	15.2	10.1	6.4	(27)	
		MCDB	10.9	16.2	21.9	25.0	27.2	29.2	29.2	29.7	29.7	24.8	20.0	13.1	(28)	
	2%	WB	2.7	5.1	9.1	11.5	16.8	20.6	21.8	20.3	18.5	13.5	7.9	3.6	(29)	
		MCDB	6.4	11.1	18.3	20.7	23.1	27.2	29.7	29.1	26.9	22.3	15.3	8.2	(30)	
	5%	WB	1.4	2.9	7.5	9.9	15.4	19.1	20.8	19.5	17.0	11.3	5.9	1.7	(31)	
		MCDB	4.1	6.2	15.7	17.5	21.9	26.2	28.8	28.0	25.3	18.3	11.8	5.4	(32)	
	10%	WB	0.4	1.2	5.8	8.3	13.9	18.0	19.9	18.5	15.6	9.7	4.2	0.3	(33)	
		MCDB	2.7	4.0	11.9	14.8	19.9	25.0	27.8	26.8	23.9	16.0	8.9	3.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	12.4	13.3	13.8	13.4	14.1	16.1	16.6	16.8	14.4	14.0	12.2	(35)	
		MCDBR	14.3	16.0	20.1	19.3	17.7	17.5	19.1	19.9	21.9	20.6	19.0	15.0	(36)	
		MCWBR	9.4	10.3	11.1	10.5	9.0	8.0	6.2	7.5	8.7	10.5	11.7	10.0	(37)	
	5% WB	MCDBR	12.5	14.7	19.6	18.7	14.9	15.6	15.4	16.5	17.2	18.4	18.8	14.4	(38)	
		MCWBR	9.4	9.7	10.8	10.1	7.8	8.2	6.8	6.9	7.6	9.8	11.5	10.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.254	0.259	0.277	0.314	0.334	0.347	0.365	0.365	0.326	0.290	0.264	0.248	(40)	
	taud		2.541	2.541	2.545	2.436	2.451	2.445	2.423	2.400	2.495	2.564	2.539	2.542	(41)	
	Ebn at Noon		887	946	967	947	933	918	896	884	896	892	863	858	(42)	
	Edh at Noon		62	75	87	107	109	110	111	110	91	73	62	56	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.79	2.80	3.92	5.01	5.90	6.78	7.03	5.94	4.57	2.97	1.95	1.46	(44)	
	RadStd		0.11	0.20	0.31	0.29	0.60	0.50	0.30	0.32	0.29	0.29	0.18	0.08	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page